

PART III

# *Core Setup*



*The multi-week project at the heart of the book — widening safety limits, installing the new guardrails, and tuning ISF weights to your meals.*

## CHAPTER 3

# General FCL Settings



**T**his chapter walks through the settings you change in AAPS Preferences before you enable autoISF's glucose-based ISF modulation. These settings widen the safety cage that HCL kept narrow, and install the new safety net (iobTH) that protects you from the wider cage.



### CRITICAL REMINDER FROM CHAPTER 1

Everything in this chapter assumes you already have a properly tuned hybrid closed loop with correct profile ISFs, and that you use a fast insulin. If not, stop and go back.

## 3.1 SMB range extension

Setting path: Preferences → OpenAPS SMB → autoISF settings → SMB delivery settings → `smb_max_range_extension`.

In AAPS Master, the maximum SMB size is 120 minutes of basal (2.0× hourly basal). In FCL, that's usually not enough — you need the loop to deliver insulin comparable in magnitude to what you used to bolus manually.

For 5-minute CGMs (Dexcom G6/G7/ONE): Start with 2.0. This doubles the limit to 4 hours of basal per SMB.

For a more precise estimate:



#### SIZING CALCULATION

*(author's numbers in parentheses)*

*Goal: within 10 minutes of a detected rise, deliver at least half of what you'd have bolused in HCL — roughly 2 SMBs worth.*

- *Target SMB size: you want at least  $(\text{your HCL meal bolus} \div 2)$  as an achievable single SMB. With a typical 8U bolus  $\rightarrow$  4U target. Allow 3U as a workable individual SMB size.*
- *Your hourly basal: e.g., 0.6U/h. AAPS Master's default 2 $\times$  allows 1.2U per SMB.*
- *Needed extension:  $3U / 1.2U = 2.5$ .*

Very low basal rates can push this number up to 3.0 or higher. The maximum setting is 5.

Keep an eye on the AUTO ISF tab or Emulator output to see whether your loop is frequently bumping against this limit. If it is, widen it.

For 1-minute CGMs (Libre 3 via Juggluco): With values every minute, insulin delivery is spread across up to 5 SMBs per 5 minutes. You rarely need large individual SMBs.

Start at 1.0 (the default). Only raise it if your basal is very low, or if your loop routinely fails to deliver small SMBs quickly enough.

## 3.2 autoISF min and max

Setting path: Preferences  $\rightarrow$  OpenAPS SMB  $\rightarrow$  autoISF settings  $\rightarrow$  `autoISF_min` and `autoISF_max`.

These cap how far autoISF can amplify or soften your profile\_ISF.

- `autoISF_max` = 2.0 to start. Up to 2 $\times$  stronger than profile\_ISF, if the weights call for it.
- `autoISF_min` = 0.3 to start. Down to 30% as strong (i.e., ISF is more than 3 $\times$  weaker).

Do not leave `autoISF_min` at 0.5 or higher — it will prevent exercise mode from properly softening the loop.

Raise `autoISF_max` later if you find the AUTO ISF tab frequently reports “capped by autoISF\_max.” Don’t raise it prophylactically.

### 3.3 SMB delivery ratio

Setting path: Preferences → OpenAPS SMB → autoISF settings → SMB delivery settings → `smb_delivery_ratio`.

Default is 0.5 — the loop delivers 50% of the currently-calculated insulin need per SMB cycle. The remaining need rolls into the next 5-minute decision.

For 5-minute CGMs: Start at 0.6 or 0.7. Higher values get you more insulin delivered 5–10 minutes earlier, which matters a lot in FCL (where you start late already).

A higher ratio also amplifies unwanted SMBs from CGM jitter. If your CGM is not rock-solid, stay near 0.5–0.6.

Do not go above ~0.8. A 100% delivery ratio works against you in two ways: - Every noisy SMB is amplified too. - You end up forced to compensate by narrowing other parameters (lower `autoISF_max`, lower `bgAccel_ISF_weight`), which makes the whole system less dynamic and costs time-in-range.

For 1-minute CGMs: Set below 0.5 (typically 0.3–0.4). The extra sampling frequency already gives you the “early delivery” benefit; a high delivery ratio on top would be too aggressive.

### 3.4 iobTH — your most important safety setting

Setting path: Preferences → OpenAPS SMB → autoISF settings → Full\_Loop\_settings → `iob_threshold_percent`.

iobTH is an iob ceiling. Once your total iob exceeds it, no more SMBs will fire. Temp basal can still run, but big insulin delivery stops. This is what keeps an aggressively-tuned FCL from stacking too much insulin.

 Everything else in autoISF settings is meaningless without a sensible iobTH.

## FIRST — SET MAX IOB

First, check that your `maxIOB` (Preferences → OpenAPS SMB → Maximum total IOB) is set correctly.

- If you have historical iob data: look at the max iob you’ve ever needed (including times of elevated insulin resistance). Set maxIOB slightly above that.
- If you don’t:

`maxIOB = hourly basal + (max meal carbs digested in ~2.5h / IC) + (correction from I`  
then multiply by ~1.2 for a cushion.

## FIND YOUR IOB TARGET

Look at your historical data. For the biggest high-carb meals you handle, what iob level (bolus + SMBs) was genuinely useful?

 *Author’s example: Big meals sometimes needed up to 8U of iob. TDD 40U. maxIOB = 10U.*

## CALCULATE IOB\_THRESHOLD\_PERCENT

Decide how much of that “big meal iob” you want autoISF to deliver rapidly via SMBs. A common choice is ~75%.

 *Example: Target iob via rapid SMBs = 75% × 8U = 6U. With maxIOB of 10U:*  
`iob_threshold_percent = 6/10 = 0.6 → 60%.`

Enter this value in AAPS Preferences.

## THE +30% OVERSHOOT RULE

The last SMB allowed can exceed the effective iobTH by up to 30%. This is on purpose: it lets iob run higher for genuinely big meals (where the final SMB is still large) without needing to set iobTH itself higher.

- At bg targets above 100 mg/dL, the overshoot is capped at +20% instead of +30%.
- An individual SMB that would overshoot by more than 30% is cut to 130% of iobTH.
- Once iob is above the effective threshold (plus overshoot), only temp basal adds more insulin.

## HIGH-CARBERS: LOWER IOBTH SLIGHTLY

If you eat big meals, keep in mind that the +30% overshoot will push your actual peak iob above the stated iobTH. Pick your `iob_threshold_percent` so that  $\text{iobTH} \times 1.3$  is the real ceiling you're comfortable with.

## IOBTH AUTO-MODULATION

iobTH is not a fixed number in daily operation. Several things modulate it:

- Low TT (e.g., an EatingSoon TT) → iobTH goes *up*, allowing more iob for the meal.
- High TT (e.g., an exercise TT) → iobTH goes *down* sharply, limiting iob for exercise.
- Exercise button (yellow, combined with TT > target) → further sharpens the reduction.
- %profile setting → iobTH scales proportionally (80% profile → 80% of iobTH).
- Activity monitor → small automatic adjustments based on step count.

All of these changes show up in the AUTO ISF tab — the “effective iobTH” is reported at the top of the results section.

## ENABLE ODD/EVEN TARGET LOGIC

In Preferences → OpenAPS SMB → autoISF settings → SMB delivery settings:

- ☐ “Enable alternative activation of SMB depending on current target” = ON.

This is required by autoISF 3.0.1+ and enables the even/odd emergency brake (§2.4).

## SETTING IOBTH WITH AUTOMATIONS

You can override iobTH via an Automation. Be careful: setting a different `iobTH_percent` via an Automation does not automatically revert when the Automation ends. You need a tandem “restore” Automation to put it back.

For this reason, prefer modulating iobTH indirectly — via TT, %profile, or the exercise mode — which revert automatically.

### 3.5 EatingSoon TT — skip for now

FCL works perfectly well without any EatingSoon TT. You can skip this section on first setup and return later if needed.

Briefly: setting a low TT (e.g., 74 mg/dL) before a meal: - Makes early SMBs slightly larger. - Raises the dynamic iobTH slightly, so SMBs keep firing for bigger meals. - Avoids halving the `bgAccel_ISF` `cap_weight` that would otherwise apply when bg sits below target.

The author's preferred approach is not to set an EatingSoon TT manually, but to use an Automation that sets a low TT automatically at the first sign of a rise (e.g., when bg delta exceeds +10 mg/dL, iob is low, meal time window applies, and no TT is already running). The duration is brief (~30 minutes).

Benefits: - First 1–2 SMBs fire before the trigger, so low-carb meals don't get over-treated. - Bigger meals automatically get the boost on SMBs 3+ and the elevated dynamic iobTH.

Drawbacks: - Random bg bumps in the time window can trigger unwanted aggressive SMBs. Tune the trigger delta carefully. - Blocks other “no TT set” Automations during that window.

Details are in §5.2 (automatic modulation). Don't worry about this on first setup.

### 3.6 Other AAPS preferences to verify

Quick checklist — the ones requiring actual attention are in bold. The rest, leave at defaults or as specified.

1. ☐ Enable SMB, SMB with high TT, SMB always, and UAM. All ON.
2. ☐ Autosens: OFF. autoISF is incompatible with Autosens in FCL mode. You'll see a warning when Autosens is on without carb inputs.
3. ☐ Autotune: do not use. Likewise, dynamicISF (in iAPS/Trio): OFF. And dynamic CR, sigmoid: OFF.
4. ☐ SMB frequency: 3 minutes (standard), 1 minute for Libre 3.
5. ☐ “High TT raises sensitivity”: ON. Required for the exercise mode math.
6. ☐ “Low TT lowers sensitivity”: OFF at first. Turn ON later, once you understand how aggressive your loop can go.
7. ☐ “Resistance lowers target” and “Sensitivity raises target”: OFF. These interact poorly with autoISF — especially the even/odd target logic.

8. ☐ Half-basal exercise target: 160 mg/dL placeholder. You'll tune this in Chapter 6.
9. ☐ "Activity modifies sensitivity": OFF until Chapter 6. (You can turn it on sooner if you're already sure about your step-count tuning.)
10. ☐ Advanced → "Always short avg delta": OFF unless your CGM is jittery. Smoothing costs you early detection time.
11. ☐ Smoothing (AAPS Configuration Builder → Smoothing): None is preferred for excellent CGMs. Average is next best. Exponential smooths nicely but hides the early rise that FCL depends on. G7 users typically need exponential smoothing.
12. ☐ Safety multipliers (Preferences → OpenAPS SMB → Advanced): roughly double the HCL defaults, so the loop can do up to 500% TBR.
13. ☐ `iob_threshold_percent` : enter the value from §3.4.
14. ☐ Password-protect Preferences with a short password, to avoid accidental changes while scrolling.
15. ☐ Disable bottom buttons (Preferences → Overview → Buttons: Insulin, Calculator, etc. all OFF). Especially important if a child uses the phone, or if it might be temporarily out of your control. The insulin button allows a direct insulin shot, which is dangerous.
16. ☐ Alarms: review your past outlier data and set xDrip / AAPS alarms that would have caught those issues.
17.  autoISF settings: do not yet enable "Enable ISF adaptation by glucose behavior." You will do this when you start Chapter 4.

### *3.7 Summary: you are ready when...*

- ☐ SMB range extension set (§3.1).
- ☐ autoISF\_max = 2.0, autoISF\_min = 0.3 (§3.2).
- ☐ SMB delivery ratio set (§3.3).
- ☐ maxIOB, iob\_threshold\_percent configured (§3.4).
- ☐ Odd/even target logic enabled (§3.4).
- ☐ AAPS preferences cleaned up (§3.6).
- ☐ Password set on Preferences.
- ☐ Bottom buttons (Insulin, Calculator) disabled.



☐ “Enable ISF adaptation by glucose behavior” still OFF.

When every box is checked, move to Chapter 4.

\* \* \*

C H A P T E R 4

# *Meals: Setting Your ISF Weights*



**T**his chapter is where you tune autoISF's five ISF weights to your meals. Each weight targets a phase of the post-meal glucose curve (§2.2). When you're done, your loop will handle your normal meal spectrum on its own.

Budget three to six weeks. Do not rush.

## *4.1 Getting started*

### RULES RECAP

1. Tune on days with normal insulin sensitivity only. No illness, no heavy exercise, no hormonal shifts. You are calibrating the baseline that the rest of the system adapts from.
2. Start with typical meals from your own spectrum. Not extremes. Not novelty meals. Not sweet drinks at meal start.
3. Iterate on 2–3 different meals, not one. Settings optimized for a single meal will rarely work for the rest. (Case Study 8.2 is the negative example.)
4. Do not give a bolus during this phase. Any user bolus distorts the glucose curve that autoISF is reading, and invalidates your tuning.

5. Start narrow. Many users begin with only one meal per day (e.g., dinners), with an Automation that turns autoISF off for the other 20 hours. Expand once one meal works.
6. Do not copy another person's numbers. Every example in this chapter is illustrative only.

#### TURN AUTOISF ON

Now — and only now — turn on Preferences → OpenAPS SMB → autoISF settings → “Enable ISF adaptation by glucose behavior”.

During your tuning period, you can automate this to be ON only during your chosen meal window (e.g., 11 AM – 6 PM for lunches) and OFF the rest of the time.

Alternatively, leave it always on and use an odd-numbered profile target for the hours you don't want it aggressive. See §5.2.

#### WATCH THE CURVES

Two things to do habitually during tuning:

1. Watch your bg, iob, and insulin activity curves develop on the AAPS home screen. Over days you'll build intuition about what “good” looks like.
2. Open the AUTO ISF tab at key moments to see exactly what the loop decided and why.

The Emulator (Chapter 11) makes this dramatically easier — you can replay a day's decisions and see everything in one place. Install it early if you're comfortable with a little Python.

#### RESIST REACHING FOR LATER TOOLS

You'll read about the DIY cockpit, Automations, exercise mode, and activity monitor. All of those are for after your weights are tuned. If you add them early, each one compensates for some weakness in your weights — and you lose the ability to tell which setting is actually doing what. Untangling that later is painful. Tune the weights first; layer the rest on top.

## 4.2 *bgAccel\_ISF — catching the first rise*

Setting: `bgAccel_ISF_weight` in Preferences → OpenAPS SMB → autoISF settings.

This is your most important weight. It drives the first 3–4 SMBs after a rise is detected — the SMBs that replace your old meal bolus.

## THE TARGET

Within about 20 minutes of acceleration detection, the first 3–4 SMBs should together deliver roughly the same job you would have bolused for this meal in HCL. In practical terms, for a big meal, you want two of those first three SMBs to each be  $\frac{1}{4}$  to  $\frac{1}{3}$  of your old HCL bolus.

- Above  $\frac{1}{3}$  per SMB is too aggressive: low-carb meals get over-treated, snacks trigger full meal responses, and CGM artefacts fire real SMBs.
- Below  $\frac{1}{4}$  per SMB is too shy: the peak runs high, and dura\_ISF has to work overtime to clean up the mess.

## STARTING POINT

Start with `bgAccel_ISF_weight = 0.02`. Yes, that small. It's a multiplier inside a formula — it adds up fast.

Test on a medium-to-high-carb meal from your normal spectrum. Don't try a pizza. Don't try a salad either. Pick something representative.

## ITERATE IN SMALL STEPS

Increase in small steps — e.g.,  $0.020 \rightarrow 0.022 \rightarrow 0.025 \rightarrow 0.030 \rightarrow 0.035 \rightarrow 0.040$  — and watch 2–3 meals over the next few days before the next change. Never change two things at once; you won't know which one caused the effect.

## CGM QUALITY IS THE MAIN VARIABLE

The main thing that distorts this tuning is CGM quality. autoISF checks a recent window of values for stability before acting on a detected acceleration.

- A jumpy or dropout-prone CGM will either (a) soften the loop's response by delaying acceleration recognition, or (b) fail to recognize it at all.
- You have three options:
  1. Use a well-performing CGM (overlapping G6 is the gold standard).
  2. Only tune on days you know are good CGM days.
  3. Live with average CGM quality and accept weaker/later SMBs and narrower overall dynamic range.

⚠ **The trap:** tuning on bad CGM days pushes you to overly aggressive weights, which then backfire on good CGM days.

## THE CAP\_WEIGHT RULE

autoISF has a built-in `cap_weight` safety factor. When bg is below target at the moment of acceleration detection, `cap_weight` halves the effective `bgAccel_ISF` contribution — the loop stays conservative when catching a rise from a low starting point. Above target, the cap doesn't apply.

This is one reason a low EatingSoon TT (§3.5) helps at meal start: it lifts your effective target above actual bg, so `cap_weight` stays at full strength and the first SMBs aren't halved.

## GOOD VS BAD TUNING

| <i>Observation</i>                          | <i>What it suggests</i>                                                            |
|---------------------------------------------|------------------------------------------------------------------------------------|
| Big meals routinely peak at 200+ mg/dL      | <code>bgAccel_ISF</code> too weak                                                  |
| First SMBs are smaller than ¼ of HCL bolus  | <code>bgAccel_ISF</code> too weak                                                  |
| Snacks and low-carb meals trigger big SMBs  | <code>bgAccel_ISF</code> too aggressive                                            |
| iob bounces over iobTH on every meal        | <code>bgAccel_ISF</code> too aggressive <i>or</i> iobTH low                        |
| Need for post-meal snacks to avoid hypos    | <code>bgAccel_ISF</code> too aggressive <i>or</i> <code>dura_ISF</code> too strong |
| Low-carb meals look fine, big ones run high | <code>bgAccel_ISF</code> probably about right, try <code>pp_</code>                |

## FIT FOR YOUR MEAL SPECTRUM

A single `bgAccel_ISF` setting has to work for:

- Low-carb meals (first SMBs must stay small)
- Medium meals
- High-carb meals (first SMBs must drive iob to iobTH fast)

This is possible, because accelerations and deltas in low-carb meals are naturally smaller than in high-carb meals, so the same weight produces smaller SMBs. Your circadian `profile_ISF` handles breakfast-vs-lunch-vs-dinner differences automatically.

If you genuinely cannot find one setting that works:

- Use an Automation to set different `bgAccel_ISF_weight` in different time windows (§5.2). Note only `bgAccel` and `iobTH_percent` are available as Automation targets.
- Use a temporary %profile switch for the brief meal window (e.g., 60% profile for 45 minutes at low-carb meals).
- Preset custom DIY cockpit buttons for snacks and special meals (§5.4).
- Accept slightly reduced performance — most users find this easier than maintaining a complex Automation web.

#### A NOTE ON DECELERATION

In the late phase of falling glucose, after the peak, the curve re-accelerates mathematically (in the downward direction). autoISF handles this internally and reduces ISF modulation as the algorithm predicts the glucose minimum. You don't need to tune anything for this phase.

### 4.3 *pp\_ISF — managing the steep linear rise*

Setting: `pp_ISF_weight`.

pp stands for “post-prandial.” This weight drives SMBs during the steep, roughly-linear rise that follows acceleration in a big-carb meal. In low-carb meals, this phase barely exists.

#### WHEN IT MATTERS

- High-carb meals, especially with sweet drinks: `bg` delta is large and sustained. If `bgAccel_ISF` hasn't already driven `iob` to the ceiling, `pp_ISF` will.
- Low-carb meals: usually skipped entirely — the curve flattens before this phase really establishes.

#### TUNING

Only tune `pp_ISF` after `bgAccel_ISF` is in a reasonable place. An over-aggressive `bgAccel` can cover up a `pp_ISF` problem, and vice versa.

- Start with `pp_ISF_weight = 0.005`. Very small.
- Test on your highest-carb meals in your normal spectrum.

- Step up by 10–20% when cautious, more when you have clear evidence.



#### THE MATH (SKIPPABLE)

*The developer's Quick Guide gives:*

$pp\_ISF = 1 + pp\_ISF\_weight \times internalFactor$

$effective\ ISF = profile\_ISF / pp\_ISF$  (when *pp* dominates)

*The `internalFactor` scales with *bg delta* above target. So *pp\_ISF* grows with the steepness of the rise.*

*Same trial-and-error principle as *bgAccel*: doubling the weight roughly doubles the amplification effect at a given *bg delta*.*

#### WHEN THE LOOP HAS ALREADY ACTED

For moderate-to-big-carb meals, the combination of *bgAccel* and *pp\_ISF* will usually drive *iob* to (or just past) your *iobTH* by the time the rise reaches its peak. Once you're above *iobTH*, only temp basal adds insulin; the rest of the weights (*bgBrake*, *dura*, *bg*) become less influential.

This is good. It means you've done the main job.

#### ONE WEIGHT OFTEN FITS ALL MEALS

Different meal types produce different delta patterns, but *pp\_ISF* scales with delta — so naturally larger deltas produce larger responses. Combined with a circadian *profile\_ISF* (different ISFs for different hours), a single *pp\_ISF\_weight* usually covers the full spectrum.

#### WATCH FOR FALSE CEILINGS

If your AUTO ISF tab frequently shows SMBs being capped by:

- `smb_max_range_extension`, or
- `autoISF_max`,

...then your *pp\_ISF* tuning is fighting a restriction instead of doing the real job. Widen the restriction first (Chapter 3) and retune.

## 4.4 *bgBrake\_ISF — the decelerating rise*

Setting: `bgBrake_ISF_weight`.

At the end of the rise, before the peak, the glucose curve decelerates. In high-carb meals, by this point your iob is usually already at iobTH and no more SMBs are firing — so `bgBrake_ISF` is irrelevant.

In low-carb meals (or with gastroparesis, or on a GLP-1 drug), the curve may decelerate before iobTH is reached, and `bgBrake_ISF` controls how much insulin continues to be delivered as the rise slows.

### STARTING POINT

Start at about half your `bgAccel_ISF_weight`. So if `bgAccel` is 0.04, start `bgBrake` at 0.02.

This reflects the reality that when bg is decelerating, insulin need drops — you usually want the loop to be softer here, not sharper.

### WHAT IT DOES

The same formula as `bgAccel` applies, but with a negative `internalFactor` during deceleration. The effective ISF comes out *bigger* than `profile_ISF` (softer), not smaller. A larger `bgBrake` weight means a softer effective ISF during deceleration.

### TUNE ONLY IF RELEVANT

For high-carb eaters, `bgBrake_ISF` rarely matters — skip the fine-tuning. Half of `bgAccel` is a reasonable default.

For low-carb eaters or slow-absorption meals, this weight plays a real role. Start at half of `bgAccel` and adjust based on whether your low-carb meals run high (reduce the weight → sharper ISF) or produce late hypos (increase the weight → softer ISF).

## 4.5 *dura\_ISF — stuck at high*

Setting: `dura_ISF_weight`.

`dura_ISF` activates when glucose plateaus at an elevated level for 10+ minutes. It boosts insulin in proportion to:

- How long the plateau has persisted



- How far above target the plateau is

This is what handles the “fatty meal second peak” or the long stuck-high plateau that high-FPU meals (pizza, burgers, cheesy pasta) produce.

#### WHEN IT MATTERS

- Fatty / high-protein meals — the classic late plateau, sometimes a second peak 2–4 hours after the meal.
- Low-carb meals — if carb absorption is slow enough to keep bg stable at mildly elevated levels for an extended period, `dura_ISF` picks up the slack.

#### STARTING POINT

Start with `dura_ISF_weight = 0.2`, and raise it cautiously. The late action of `dura_ISF` means a weight set too high will hypo you 2–3 hours after a meal, when you’re not paying attention.

#### PROTECTING AGAINST LATE DURA HYPOS

`dura_ISF` keeps working the whole time a plateau lasts. That’s useful for a genuine fatty-meal plateau — but dangerous if the plateau is actually an insulin tail fading out. Two Automation patterns help:

- Fade dura out as the plateau ages. Set a low TT near the actual target when a high plateau is clearly meal-related. This lets dura act sharpest at the start of a persistent high, then taper off.
- Block dura below ~140 mg/dL. An Automation that shuts off SMBs when a mild plateau forms below that threshold. Late insulin at that level rarely pays off; hypo risk does.

These are refinements you add after your weights are stable. The `04_ISF_weights_effects` spreadsheet (in the book’s repo) lets you model how your specific `dura_ISF` setting responds before you commit to a pattern.

Start with the default weight and tune conservatively first.

#### WHAT GOOD TUNING LOOKS LIKE

- Persistent highs resolve over 1–2 hours, not 4+.
- No late hypos (2–4 hours post-meal).

- `dura_ISF` shows contributions in the AUTO ISF tab during plateaus but isn't the dominant factor during acute rises.

#### IF YOU SEE LATE HYPOS

In order of preference: 1. Fix `bgAccel_ISF` and `pp_ISF` first. If the initial rise is handled better, the peak is lower and the plateau shorter — `dura_ISF` doesn't need to work as hard. 2. Lower `dura_ISF_weight` in small steps. 3. Add an Automation to cap `dura` contributions once `iob` exceeds a threshold — advanced, see Chapter 5.

## 4.6 *bg\_ISF — generally not needed in FCL*

Setting: `bg_ISF_weight` (under Preferences → OpenAPS SMB → autoISF settings → `bg_ISF` settings).

`bg_ISF` strengthens ISF at high `bg` and softens it at low `bg`, independent of meal phase. In HCL it can be useful; in FCL it mostly isn't.

#### WHY FCL SKIPS IT

In FCL you already get most of your insulin early, through `bgAccel` and `pp_ISF`, when `bg` is still relatively low. Boosting ISF again at high `bg` risks over-delivery; softening at low `bg` reduces the early SMBs you rely on.

#### RECOMMENDED FCL SETTINGS

- `lower_ISF_range_weight = 0.0`
- `higher_ISF_range_weight = 0.0`

If you have specific evidence that you'd benefit from mild modulation (e.g., you reliably run slightly below target after correcting a high), try 0.1–0.2 on the relevant side and analyze.

## 4.7 *How the loop copes without carb entry*

Since you give no carb inputs in FCL, the `oref UAM` (Unannounced Meal) algorithm has to estimate carbs on its own. It works in two directions:

- Looking back — given the observed `bg` change and your profile ISF/IC, it calculates how many grams “must have been” absorbed in the last 5-minute window.

- Looking forward — it projects continued absorption for the next 1–3 hours, based on recent “carb deviation” data.

The prediction isn’t perfect, but it’s often no worse than a user’s eCarb estimate — and it’s recalculated against reality every 5 minutes. The practical consequence: you don’t need to know how many grams were on the plate.

FCL still has to manage the late-meal phase carefully — fading insulin tail meets uncertain remaining carbs. That’s what `dura_ISF` and a well-set `iobTH` are there for.

## 4.8 Iterating across meals

Take 2–3 meals from your normal spectrum. For each:

1. Observe the full ~3 hours after the meal.
2. Note where the loop did well and where it fell short (too-high peak? late hypo? over-reacted to a snack?).
3. Pick the *first* thing to change — usually the earliest-acting weight with a problem.
4. Make a ~10–20% change.
5. Repeat for a few more meals before changing again.



THE TEMPTATION TO OPTIMIZE FOR ONE MEAL IS THE SINGLE BIGGEST MISTAKE IN THIS PROJECT.

*A weight set that handles your pizza perfectly will probably fail on your salad. Iterate across meal types, always.*

### QUALITY CHECK — THE AUTO ISF TAB

At key moments during a meal, open the AUTO ISF tab. Check:

- Which ISF component (`acce` / `pp` / `brake` / `dura` / `bg`) is dominating `final ISF factor`?
- Is the resulting `sens` within the `autoISFmax` / `autoISFmin` bounds, or bouncing against a limit?
- Is the SMB size capped by `smb_max_range_extension` or `autoISF_max`?
- Is `bg` below target, triggering the `cap_weight` halving?

- Is iob above iobTH, triggering SMB shutoff?

Any of these can mask a setting you meant to tune. Fix the masking issue before continuing.

#### MEALS THAT SIMPLY WON'T FIT

For unusual meals that resist any reasonable fit:

1. Accept slightly worse time-in-range for those meals and move on.
2. Keep a small snack on hand for when the insulin tail drops you after a high peak.
3. Manually intervene — a brief odd TT for a snack situation, for example.
4. Define a User Action Automation that gives that meal type its own settings. See §5.4.
5. Fall back to HCL for those specific meals.

## 4.9 Focus your effort



#### PRIORITY ORDER FOR YOUR WEIGHT TUNING:

1. *bgAccel\_ISF\_weight* — by far the most important. Spend the most time here.
2. *iobTH\_percent* — revisit after *bgAccel* is stable.
3. *pp\_ISF\_weight* — second most impactful.
4. *dura\_ISF\_weight* — mostly for low-carb or fatty meals.
5. *bgBrake\_ISF\_weight* — leave at half of *bgAccel* unless low-carb is a big part of your diet.
6. *bg\_ISF\_weight* — leave at 0 in FCL.

Once you've found settings that work across your normal spectrum, don't keep tuning. Chapter 9 (Performance Monitoring) covers when re-tuning is actually warranted — almost always, the answer is “not now.”

## *4.10 Complex scenarios — where they're covered*

Everything beyond “my normal weekday meals” is handled in later chapters:

- Snacks, sweet drinks, out-of-spectrum meals → Chapter 5 (modulation).
- Exercise before, during, or after a meal → Chapter 6.
- Illness, hormonal shifts, steroid days → Chapter 5 (%profile switch).
- Kids → Chapter 7 (Meal Announcement mode is often better).

Finish tuning for your normal spectrum first. Then layer the rest on top.

\* \* \*

PART IV

# *Day-to-Day Operations*



*Nudging the loop when it needs nudging — snacks, exercise, sick days, and everything else that isn't a normal meal.*

CHAPTER 5

# Modulating Loop Aggressiveness



Once Chapter 4 is done, your FCL handles normal meals hands-off. This chapter is about everything else — snacks, sick days, rough CGM days, diet deviations, nights, compression lows, and all the rest of ordinary life.

You have five levers for adjusting loop aggressiveness. Some are automatic, some manual, some you pre-program into buttons on your AAPS home screen.

## 5.1 The five levers

| <i>Lever</i>           | <i>What it does</i>                                       |
|------------------------|-----------------------------------------------------------|
| Odd-numbered target    | Shuts SMBs off entirely (§2.4). Emergency brake.          |
| Change ISF weights     | Shifts the acceleration/duration behavior.                |
| Change iobTH_percent   | Raises or lowers the iob ceiling.                         |
| Change %profile        | Multiplies everything (basal, ISF, iobTH) proportionally. |
| Set a temp target (TT) | Low TT → more aggressive; high TT + exercise → less.      |

All five are also accessible in AAPS Automations, which means any pattern you can describe can be triggered automatically when your conditions are met.

💡 *The core discipline: keep it simple. A loop that works because of five Automations interacting with each other is a loop nobody — including you — understands. Do the basics well; add Automations sparingly.*

## 5.2 Fully automatic modulation

This section is about configurations that run without user action. Use these for recurring, predictable situations.

### DISABLE OUTSIDE MEAL WINDOWS

Some users keep autoISF active only during the day (or only at meal times), and run vanilla HCL — without ISF adaptation — overnight.

Two ways to do it:

- Automation on: define time windows when “Enable ISF adaptation by glucose behavior” turns ON; rest of the day it’s off.
- Automation off: conversely, turn it off at bedtime and on in the morning.

This is a common starting pattern — especially valuable in your first weeks, when you want aggressive FCL for dinners but peace of mind overnight.

### ODD TARGETS DURING QUIET HOURS

A simpler alternative: set an odd-numbered profile target for the hours you want SMBs blocked (e.g., night hours).

- autoISF keeps running.
- Temp basal still adjusts insulin delivery dynamically.
- But no SMBs fire during the odd-target hours.

This is the author’s preferred night-time solution, especially for users who get compression lows at night. The loop stays “awake” enough to handle light corrections but can’t overreact to a bg jump caused by rolling onto the sensor.



💡 *Allowing a few SMBs anyway — e.g., after a late fatty meal that might plateau high. Define an Automation that temporarily sets an even target under specific conditions (bg high, iob low, late at night). Use short durations; the same Automation will retrigger if conditions persist.*

## DIFFERENT SETTINGS BY TIME SLOT

You can differentiate by time of day via Automations that set different values for:

- `iob_threshold_percent`
- `bgAccel_ISF_weight`
- Enable/disable ISF adaptation
- (plus TT and %profile, always)

Example use cases:

- Breakfast at home: standard settings.
- School lunches: milder bgAccel to avoid over-treating a quick cafeteria snack.
- Dinners at home: highest iobTH for big meals.

⚠️ **Most users don't need this.** Unless your meals differ dramatically by time slot, the circadian `profile_ISF` alone (your hourly ISF schedule) handles the natural differences between breakfast, lunch, and dinner. Try to tune one broader spectrum first (Chapter 4). Only differentiate if that genuinely fails.

## ACTIVITY MONITOR

The Activity Monitor uses your phone's step counter to auto-adjust insulin sensitivity for recent (last ~60 minutes) activity levels. See §6.6 for tuning details.

Enabling it in Preferences → OpenAPS SMB → “Activity modifies sensitivity” gives you automatic hands-off handling of things like “I walked the dog after lunch” or “I’ve been on the couch all afternoon.” It’s capped at +20% insulin (for detected inactivity) and <sup>&</sup> 30% (for activity).

For users whose daily variability is mostly step-count driven, this single feature can close most of the gap toward truly 24/7 hands-off FCL.

## HOW HANDS-OFF CAN FCL REALLY BE?

Fully hands-off 24/7 FCL is achievable for adults with:

- Stable technical setup (cannula, CGM, Bluetooth).
- Moderate and relatively regular meals.
- Moderate %TIR expectations (upper 70s to low 80s).
- Some willingness to use an Automation or two for recurring situations.

It's less realistic for users with highly varied meals, heavy sports, or extreme %TIR goals. For those, expect to do some nudging (§5.3–7.4).

## 5.3 Manual modulation — the “FCL cockpit”

Your AAPS home screen has three buttons at the top: %profile, Exercise, TT. In FCL these are your cockpit — the fastest way to adjust loop behavior without going into Preferences.

### THE THREE TOP BUTTONS

| Button   | Grey means     | Yellow means                                                        |
|----------|----------------|---------------------------------------------------------------------|
| %profile | 100% profile   | Temporary %profile active (<100% softer, >100% sharper)             |
| Exercise | Off            | On — dynamic exercise mode armed (needs elevated TT to activate)    |
| TT       | No temp target | Temp target active (color also hints at above/below profile target) |

All three buttons yellow ( YYY or yyy depending on direction) is the “exercise mode with modified profile” combination. All three grey ( GGG ) is default profile operation.



## THE FULL COLOR-COMBINATION TABLE

*(read only if you're curious)*

Lowercase **y** means softer than default; uppercase **Y** means more aggressive. Position matters: **%profile / exercise / TT**. So **YGY** = elevated profile × no exercise × elevated TT (unusual), **yyy** = softened profile × exercise × elevated TT (classic exercise combination).

Most users only use three real combinations: - **GGG** — default. - **yyy** or **Gyy** — exercise with TT, optionally softened profile. - **GGY** — lowered TT (e.g., *EatingSoonTT*) for more aggressive response.

The remaining combinations exist for special cases. Don't memorize them; look them up if needed.

## QUICK MODE CHANGES

Temporary %profile. A single number that scales everything.

- 110% for an afternoon when you feel resistant → slightly sharper everywhere.
- 80% for the evening after a heavy exercise day → slightly softer everywhere.
- Safety limits (Chapter 3) must be wide enough that a 110% doesn't get cut off.

Temporary target (TT). Use this freely:

- Low TT (e.g., 74 mg/dL) → loop aims for a lower target and applies sharper ISF. Common “EatingSoon” usage.
- High TT → loop aims higher and softens ISF. Common for exercise.
- Even vs odd target — remember, odd numbers block SMBs.

Exercise button. Toggle with the middle top button. When on *and* a TT above profile target is also set, the dynamic exercise mode activates — see Chapter 6 for the math.

## TEMPORARY SETTINGS IN PREFERENCES

Sometimes you want to change an **ISF\_weight** or **iobTH\_percent** value directly. You *can* — go into Preferences, enter your password, make the change. But this route has real problems:

1. It requires multiple taps and a password.

2. You will forget to change it back.

For this reason, use the three top buttons (TT / %profile / exercise) for temporary adjustments. They revert on their own. Save Preferences edits for permanent tuning changes.

## 5.4 The “DIY cockpit” — custom buttons from User Action Automations

You can add your own buttons to the bottom of the AAPS home screen by defining an Automation with the “User action” checkbox ticked. The button appears only during the time window you configure, and only while any other conditions you set remain true.

This is how you pre-program responses to recurring situations.

### A SNACK BUTTON

Suppose you sometimes have a 15g fast-carb snack in the afternoon. You don’t want the loop to treat it like a full meal:

1. Create an Automation named “Snack 15g fast-C.”
2. Set User action = ticked. Configure a time window (e.g., 2 PM – 6 PM) when the button should appear.
3. Under Actions:
  - Set a low TT (e.g., 74 mg/dL) for ~20 minutes (sharpen the initial SMBs).
  - Raise `bgAccel_ISF_weight` briefly (or set a temporarily higher %profile).
  - Lower `iob_threshold_percent` to prevent over-insulinization (e.g., to 35%).
4. Save.

Now, any afternoon you’re about to snack, you tap the button and your loop will handle the snack differently.



**CRITICAL: IOBTH AND BGACCEL\_ISF\_WEIGHT DO NOT AUTO-REVERT.**

Setting a low `iob_threshold_percent` via an Automation permanently shifts it until something else puts it back. You need a tandem Automation — see §5.5.



**Don't press the snack button twice.** The lowered `iobTH` won't allow enough `iob` for a full meal. If you find yourself snacking more than planned, just let the regular meal response run instead.

## OTHER COCKPIT BUTTON USES

- Exercise announcement — see §6.4.
- Meal-before-exercise — the tricky scenario in §6.5.
- Sweet drink — like the snack, but with different settings.
- “Pause FCL” — briefly disable ISF adaptation, for a difficult situation.

## MANAGING YOUR AUTOMATION LIST

Over time your list grows. Discipline:

- Shelf unused Automations — untick the top-left checkbox to make them inactive. They stay in the list for reactivation but don't execute.
- Review periodically — if you haven't used an Automation in months, remove it.
- Make evening a review habit — before bed, glance at what's active for tomorrow. Turn on what you'll need; shelf what you won't.

## *5.5 The tandem Automation pattern — critical*

Many Automation Actions (`iobTH_percent`, `bgAccel_ISF_weight`, ISF adaptation on/off) cannot be set with a duration. Once your Automation runs, that setting stays changed forever, until something else restores it.

The solution: a second Automation, running in tandem, whose job is to restore the default value when the first Automation's conditions are no longer met.

## HOW TO STRUCTURE IT

Imagine you have several Automations that might lower your `iob_threshold_percent` (snack mode: 35%, exercise-after-meal: 40%, etc.). Your default is 60%.

Create one restore Automation:

- Conditions: `iob_threshold_percent > 45%` (or whatever's higher than any lowered value), AND no TT active, AND time window is back to normal.
- Action: Set `iob_threshold_percent = 60%`.

This runs after any of your lowering Automations time out.

## AUTOMATION INTERACTIONS

A set TT often blocks other Automations from running. This is by design — once you're in a “special mode,” you don't want random Automations overriding your manual choice. Consequences:

- Short TT durations are better. If your conditions still apply, the Automation will simply re-trigger.
- Sequence matters — if you have many Automations, the loop evaluates them in order.
- If you need an Automation to run *during* a TT, think carefully about whether that's right, and build a test for it.

 *Don't build loops inside your loop. If you find yourself writing Automations that rapidly set low TTs, then cancel them, then trigger other Automations based on bg deltas — stop. You're likely duplicating what autoISF already does internally, only 5 minutes late. Fix the underlying weight instead.*

## 5.6 Temporarily exiting FCL

Sometimes the best move is to stop FCL entirely and go back to manual meal bolusing.

### THREE WAYS TO EXIT

1. Switch off ISF adaptation. Preferences → OpenAPS SMB → autoISF settings → “Enable ISF adaptation by glucose behavior” = OFF. Fastest complete shutdown.
2. Temporarily set `bgAccel_ISF_weight` to 0. Keeps `dura_ISF` and the rest working, but disables the aggressive first-SMB behavior.

3. Tap the loop icon (violet circle → green circle) if your AAPS version includes this UI feature. This will switch the Overview buttons back on too.

After exiting, you need to remember to:

- Re-enable the insulin button (Preferences → Overview → Buttons → Insulin) to manually bolus.
- Bolus for meals yourself.
- Re-enable FCL when ready — nothing does this automatically.

#### A “PAUSE FCL” BUTTON

Because forgetting to re-enable is common, consider a User Action Automation with:

- Conditions: User action, time window, short duration.
- Action: Set `bgAccel_ISF_weight` = 0.
- Tandem restore Automation: restores the weight to your tuned default after the time window.

This gives you a fast pause that automatically ends.

 **Test it.** The User Action Automation feature for time-limited FCL pause is newer. Verify yours reverts correctly before you rely on it.

## 5.7 Recognizing the current loop state

At any moment, four places tell you what your loop is actually doing:

1. The three top buttons (§5.3) — yellow or grey.
2. The `ai: %` indicator under Autosens% on the home screen — shows whether autoISF is adapting right now.
3. The AUTO ISF tab — full detail on the most recent loop decision, including:
  - Starting profile\_ISF (including any Activity Monitor or TT adjustment).
  - Each component’s contribution (acce / pp / bg / dura).
  - Final effective ISF ( `sens` ).

- Effective iobTH in use.
  - Whether any safety limit capped the SMB.
4. Optional extra graphs below the main AAPS glucose chart — iobTH corridor, ISF factor contributions, fitted parabola.

Between these, you should always be able to answer “why did the loop do that?”

## 5.8 Managing snacks — a worked example

Snacks are the most common situation where users need modulation. A summary of your options:

| <i>Snack type</i>                       | <i>Best handling</i>                               |
|-----------------------------------------|----------------------------------------------------|
| Small anti-hypo snack (glucose tabs)    | Odd TT for ~30 min to block SMBs.                  |
| Small, slow, low-carb snack             | Let FCL handle it. Check iob to be sure.           |
| 15–30g fast-carb fun snack (ice cream)  | DIY cockpit button — low iobTH, sharper weights.   |
| Multi-snack situation (evening grazing) | Set 80–90% %profile for a few hours + mindfulness. |
| Mid-sports snack                        | See Chapter 6. Usually odd TT + no special action. |

Case Study 5.2 (Sweet snacks / Glühwein with DIY cockpit) gives a detailed worked example.

## 5.9 Summary

- Five levers, three buttons, and a handful of well-designed Automations handle nearly every non-meal situation.
- Keep Automations few and simple. Always pair aggression-changing Automations with tandem restore Automations.
- Use odd TT as your emergency brake.
- Pause FCL if things feel wrong — re-enable when ready.
- Consult the AUTO ISF tab any time you’re confused about what the loop is doing.



\* \* \*

C H A P T E R 6

# *Exercise and Activity*



**E**xercise is the single biggest source of unplanned hypoglycemia in any looping setup. This chapter covers the tools autoISF gives you for it.

The key insight: you cannot prepare for exercise the way you did in HCL, because FCL is tuned to react to rising bg with aggressive insulin. Setting a pre-exercise high TT before a meal would sabotage the meal response. Setting one after the meal starts is too late — the aggressive insulin is already on board.

autoISF gives you three tools to navigate this:

1. Dynamic iobTH — the iob ceiling auto-adjusts based on TT and exercise button.
2. Temporary %profile switch — scales everything down.
3. The “DIY cockpit” + Automation sequence — pre-programs the whole thing into one button press.

## *6.1 You don't need every tool here*

Skim the sections below, then focus on what applies to you:

- Light exercise only (dog walks, gardening, light housework) — the Activity Monitor (§6.6) probably handles you. Skip to that.
- Regular specific exercise (biking, gym, hikes) — you'll use Dynamic iobTH (§6.2) and the DIY cockpit (§6.4).

- Occasional unusual exercise — the cockpit button (§6.4) can be pre-configured and then activated only on those days.

You can also always drop back into HCL and handle exercise the traditional way while you're setting up FCL.

At the bare minimum, keep snacks on hand. That is always the fallback.

## 6.2 Dynamic iobTH — the main exercise tool

Dynamic iobTH is how autoISF recognizes exercise: when both the exercise button is on *and* the active TT is above profile target, the iob ceiling drops automatically. No extra Automation needed.

Two inputs control how aggressive the drop is:

- `half_basal_exercise_target` in Preferences → OpenAPS SMB. Default 160 mg/dL. Lower values produce more aggressive softening.
- The TT you set for the specific exercise (plus the exercise button being on).

### WHAT HAPPENS DURING EXERCISE

When exercise mode is active (exercise button yellow + TT above profile target), everything scales by a single factor called `sens.ratio`:

- `temp basal = profile basal × sens.ratio`
- `temp ISF = profile ISF / sens.ratio` (softer, i.e., higher)
- `temp iobTH = set iobTH × sens.ratio`

A `sens.ratio` of 0.5 means half the basal, half the iobTH, and double the ISF for the duration.

### THE MATH

The formula is `sens.ratio = (HBET - 99) / ((HBET - 99) + (TT - 99))`, where HBET is your half-basal exercise target. Picking your TT and HBET together picks your `sens.ratio`:

| <i>half-basal target</i> → | <i>180</i> | <i>160 (default)</i> | <i>140</i> | <i>120</i> |
|----------------------------|------------|----------------------|------------|------------|
| TT = 100                   | 1.00       | 1.00                 | 1.00       | 1.00       |
| TT = 120                   | 0.79       | 0.74                 | 0.66       | 0.50       |
| TT = 140                   | 0.66       | 0.60                 | 0.50       | 0.34       |
| TT = 160                   | 0.57       | 0.50                 | 0.40       | 0.26       |
| TT = 180                   | 0.50       | 0.43                 | 0.34       | 0.21       |

(The constant 99 is AAPS’s internal “normalTarget.” It changed from 100 to 99 in AAPS 4.2.1 to accommodate mmol/L users; the effect on `sens.ratio` is small.)

#### WORKED EXAMPLE

Your settings: iobMAX 10U, iobTH 60% (= 6U), profile ISF 39 mg/dL/U, half-basal target = 160 (default).

Scenario: exercise button on, TT = 160 mg/dL.

- `sens.ratio = (160-99) / ((160-99) + (160-99)) = 0.50`.
- Temp iobTH = 6U × 0.50 = 3.0U. Temp ISF = 39 / 0.50 = 78 mg/dL/U (softer).

Add a 70% %profile on top and the effects multiply: temp iobTH = 3.0 × 0.7 = 2.1U. That’s about a third of your normal insulin ceiling — a meaningful guardrail during exercise.

#### WHAT TO LOOK UP

You don’t have to compute any of this. Open the AUTO ISF tab (or the graph that plots iobTH over time) and you’ll see:

- Profile ISF.
- Modifying factors currently active (Activity Monitor, %profile, TT, exercise mode).
- Effective ISF in use.
- Effective iobTH in use.

Every time you change a TT or the exercise button, glance at the AUTO ISF tab to confirm the loop is in the state you intended.

## TUNING THE EXERCISE SETUP

Personalize step by step:

1. Set `half_basal_exercise_target` in Preferences. Lower number = more dynamic range. Start at the default 160; lower only if you need more softening at moderate TTs.
2. For each kind of exercise you do regularly, find a TT that produces the right amount of softening. Mnemonic: choose a memorable number so “TT = 140 = my jogging TT” is easy to remember.
3. Iterate across a few sessions.

## EVEN VS ODD EXERCISE TT

An odd-numbered TT (e.g., 141, 159) blocks SMBs entirely; an even TT (140, 160) allows them but softened. Which you pick matters:

| Choice  | Behavior                                                       | Use when...                                                   |
|---------|----------------------------------------------------------------|---------------------------------------------------------------|
| Odd TT  | No SMBs at all during exercise                                 | Calm, steady activity; want zero chance of SMB interfering    |
| Even TT | SMBs allowed, but softened via reduced <code>sens.ratio</code> | Adrenaline-heavy sports; occasional spikes you’d like treated |

Odd TT is simpler and often the right default. It does have downsides: once set, any Automation that requires “no TT active” won’t run; and a brief automatic SMB (for an unexpected spike) is impossible. Know your exercise pattern.

## MANUAL IOBTH (RARELY NEEDED)

You *can* change `iob_threshold_percent` directly in Preferences for exercise. The author doesn’t recommend it:

- Multiple taps + password.
- You will forget to change it back.
- The dynamic mechanism above does the same job automatically.

Skip this route unless you have a specific reason.

## 6.3 Temporary %profile switch

The %profile switch (top-left button on the AAPS home screen) multiplies basal, ISF, and iobTH by a fixed percentage. Easy, complementary to the TT/exercise mechanism.

Common uses:

- Before/during/after exercise: 70–80% profile for a few hours to days, to cover “long-waved” sensitivity increase.
- During the day after a long sports day: 80% profile.
- Brief meal-specific nudge: 60% profile for 45 minutes at a low-carb meal.

%profile effects multiply with the exercise-mode sens.ratio. You can stack both for maximum softening during high-intensity exercise.

Caveat: a reduced %profile also reduces max SMB size (because it shrinks basal). At exercise this is usually desired, because you don’t want big SMBs while already running soft.

## 6.4 The DIY cockpit for exercise

For exercise you do regularly, prepare a custom button. Click once, get the full setup.

### A SIMPLE EXERCISE BUTTON

User Action Automation:

- Conditions: User action ticked; time window when you might exercise (e.g., 4 PM – 8 PM if you bike after work).
- Actions:
  - Set an exercise TT (e.g., 140 mg/dL) for 1–2 hours.
  - Turn the exercise button on (yellow).
  - Set %profile = 80% for 2 hours.

Tap the button when you’re about to start exercise. Everything above activates.

## DYNAMIC VS TRADITIONAL MODE

Two ways to soften the loop for exercise:

- Traditional mode — %profile reduced and a TT set, but the exercise button stays grey. You get manual softening from the %profile, with the TT as an extra lever. No dynamic `sens.ratio`.
- Dynamic mode — exercise button on (yellow). The `sens.ratio` from §6.2 applies, giving much stronger softening at moderate TTs.

Use dynamic mode. It's more precise, and you tune it through one parameter (the half-basal exercise target) instead of guessing profile percentages.

## WHEN TO SET IT UP

Before your next planned exercise. Even a rough first version is better than nothing. Refine after each use.

## *6.5 Exercise after a meal — the tricky one*

This is the hardest scenario in FCL and deserves careful handling.

### THE PROBLEM

In HCL you'd reduce your meal bolus. In FCL you don't give a meal bolus — but you also can't pre-set an exercise TT, because that would sabotage the aggressive initial response to the meal.

You want full meal response for the initial rise, capped early so exercise doesn't have too much insulin working.

### MANUAL APPROACH

1. Before the meal: lower `iob_threshold_percent` (e.g., from 60% to 40%). Keep everything else normal.
2. After iobTH is exceeded (approximately when the first meal SMBs finish): set an elevated TT and the exercise mode.

The first step caps how much iob accumulates. The second kicks in the dynamic exercise behavior.

The problem: two interventions, one of them time-critical. You'll forget or mistime it.

## AUTOMATED 3-STEP SEQUENCE

This is the recommended pattern. Three Automations, only one of which you trigger manually.

Automation #1 — “Meal before exercise” (User Action): - Conditions: User action ticked; meal-window time slot; no TT active. - Actions: Lower `iob_threshold_percent` to your exercise value (e.g., 40%). Optionally raise `bgAccel_ISF_weight` briefly (full aggressive initial response). - Tap this button before your pre-exercise meal starts.

Automation #2 — “Meal served — switch to exercise mode” (Automatic): - Conditions: iob has exceeded the lowered iobTH (or a proxy signal: iob > 4U, the elevated bgAccel flag still set, etc.). - Actions: Set %profile = 70%. End any active TT — a new one has to be set fresh in #3.

Automation #3 — “Activate exercise TT” (Automatic): - Conditions: %profile just switched to 70% and no TT is currently active. - Actions: Set the exercise TT (e.g., 125 mg/dL) for the exercise duration.

Now one button tap at meal start handles: 1. Full meal response. 2. Early iob cap. 3. Transition to exercise mode.

See Case Study 6.2 for a detailed worked example.

## THE LAISSEZ-FAIRE OPTION

Simpler but less optimal:

- Set an elevated TT + exercise mode before the meal.
- Accept that the meal peak will run higher than usual.
- Especially works if the meal is protein/fat-heavy (slow absorption), or if you’re OK starting exercise at a higher bg.

Logic: for fast-carb meals you want FCL aggressive up to a lowered iob cap, like a reduced HCL bolus. For slow meals at modest bg peaks, pre-softened FCL is perfectly fine.

Choose based on your own pattern.

## 6.6 Activity Monitor — step-counter-driven adjustment

The Activity Monitor uses your phone’s step counter to quietly nudge insulin sensitivity based on recent activity — no buttons, no Automations, just a small continuous adjustment.



## WHEN IT RUNS

- Enabled in Preferences → OpenAPS SMB → “Activity modifies sensitivity” + set scaling factors.
- Active whenever no TT is running.
- Pauses silently when any TT is set (so exercise mode takes over cleanly). Resumes the moment the TT expires.

## WHAT IT ADJUSTS

Step count over the last ~60 minutes → small automatic adjustment of:

- Basal
- ISF
- iobTH

All proportional. All capped: at most +20% insulin for detected inactivity, at most <sup>&</sup> 30% for detected activity.

## WHY USE IT

If your daily sensitivity variation is mostly driven by “desk day vs walking day,” the Activity Monitor handles it automatically in a ~60-minute feedback loop. This is much faster than Autosens (8–24 hours) and is one of the things that makes hands-off FCL feasible.

## TUNING THE SCALING FACTORS

Set scaling factors in Preferences. You personalize them to how your own body responds to gentle activity. The autoISF Quick Guide (page 9) covers the mechanics.

Rough approach: 1. Turn Activity Monitor on with starting factors. 2. On a few typical days (some walking-heavy, some sedentary), check the AUTO ISF tab to see what it’s doing. 3. If it over-softens during light walks, reduce the activity factor. 4. If it doesn’t soften enough, increase the activity factor. 5. Same for inactivity.

## A SIDE BENEFIT FOR SNACKS

If you sometimes use brief even/odd target switches (e.g., to sneak in a small snack without an SMB), the Activity Monitor resumes its work immediately after the TT ends. You don’t lose your step data.

## 6.7 Summary

- Dynamic iobTH + exercise TT + half-basal exercise target are the core exercise tools.
- The 3-Automation pattern handles the “meal before exercise” problem with one button tap.
- The Activity Monitor handles ordinary daily variation automatically.
- Keep odd-TT as your emergency SMB shutoff during surprise exercise.
- Snacks on hand are always the fallback. Never trust any configuration enough to skip that.

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